

PROJECT AND TEAM INFORMATION

Project Title

(Try to choose a catchy title. Max 20 words).

RescueRoute – Disaster Management and Evacuation System

Student / Team Information

<i>*Team Name:*</i> <i>*Team #*</i>	<i>*Crisis Coders*</i> <i>*T138*</i>
Team member 1 (Team Lead) (Last Name, name: student ID: email, picture):	Jindal, Kavya – 2510012913 2510012913geu.ac.in 
Team member 2 (Last Name, name: student ID: email, picture):	Kalra, Khushi – 2510380040 2510380040geu.ac.in 
Team member 3 (Last Name, name: student ID: email, picture):	Shukla, Bhoomi – 2510010403 2510010403geu.ac.in 
Team member 4 (Last Name, name: student ID: email, picture):	Agarwal, Ketan – 2510370403 2510370403geu.ac.in 

PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

(Describe the problem and why it is important. Max 300 words).

Disasters like floods, earthquakes, fires and landslides can require quick evacuation. Blocked roads make route planning difficult. Emergency teams must coordinate evacuees, ambulances, patients and hospitals. Critical patients need priority, while hospital capacity is limited. The proposed system provides a computerized disaster response and evacuation solution using Data Structures and OOP.

State of the Art / Current solution (1 pt)

(Describe how the problem is solved today. Max 200 words).

Modern evacuation planning uses GPS, GIS, road networks and traffic information. This project represents roads as a graph using an adjacency list. BFS finds available routes while avoiding blocked roads. Queues and priority queues manage evacuation and emergency requests, while C++ classes represent patients, ambulances and hospitals.

Project Goals and Milestones (2 pts)

(Describe the project goals and milestones. Max 300 words).

The goal is to build a disaster-response and evacuation system using Data Structures in C and OOP in C++. The system will use Graph and BFS for routes, Queue and Priority Queue for requests and triage, Linked List for records, Stack for history, Searching and Sorting for management, and File Handling. C++ classes will represent Patient, Ambulance, Hospital, Disaster and Evacuee. Milestones include developing data structures, BFS, C++ classes, integration, emergency modules, simulation, file handling and testing.

Project Approach (3 pts)

(Describe the solution, platforms and technologies. Max 300 words).

The project will be a console-based C/C++ application. C will implement Graph, Queue, Priority Queue, Linked List, Stack, BFS, Searching and Sorting. C++ classes will represent Patient, Ambulance, Hospital, Disaster and Evacuee. BFS will find safe routes, while priority queues manage emergency cases. File handling will store records and reports.

System Architecture (High Level Diagram)(2 pts)

(Provide an overview of the system and its main components in a diagram using a tool of your choice).

C++ Application → Management Layer → Data Structures → Algorithms
Patient – Ambulance – Hospital – Disaster
Graph – Queue – Priority Queue – Linked List – Stack
BFS – Searching – Sorting – File Handling

Project Outcome / Deliverables (1 pts)

(Describe the project outcomes / deliverables. Max 200 words).

A working console-based Disaster Response and Evacuation Management System demonstrating Graph, BFS, Queue, Priority Queue, Linked List, Stack, Searching, Sorting and File Handling. The system will include C++ classes for Patient, Ambulance, Hospital, Disaster and Evacuee. The simulation will demonstrate disaster registration, route finding, triage, ambulance dispatch, hospital allocation and reporting.

Assumptions

(Describe the assumptions used to solve the problem. Max 100 words)

The system is a small-scale academic simulation using an unweighted road graph. Roads may have LOW, MEDIUM, HIGH or BLOCKED status. BFS is used for routing. Hospital capacity, ambulance availability and patient severity are stored as records. Real-time GPS, GIS, cloud services, advanced traffic models and Dijkstra are outside the project scope.

References

(Provide resources or references used for this deliverable).

- 1. Kazi Ashik Islam et al., "A Simulation-based Approach for Large-scale Evacuation Planning," IEEE, 2020.*
- 2. Cormen et al., Introduction to Algorithms, MIT Press.*
- 3. Bjarne Stroustrup, The C++ Programming Language.*
- 4. Disaster Response & Evacuation Management System – PBL Project Specification.*